

Indian Institute of Technology (Indian School of Mines) Dhanbad

Department of Mining Engineering

Mid-Semester Examination (Winter Session) 2024-25 for B.Tech. Students (103 nos.)

Subject: Advanced Underground Mining Methods (MND 411) Venue: NLHC LH 04-06

Time: 120 Minutes (03.00 PM-05.00 PM) Date: 21.02.2025 Maximum Marks: 60

Questions	Section A- 30 Marks Answer any three questions from Questions 1 and 4 [3×10 = 30 Marks]	Marks
1.	Discuss the operational challenges associated with thick seam mining. Enumerate the typical ground control requirements for mining thick seams. (Schematics are required)	[10]
2.	a. What are the different zones that form above the coal seam with the progressive mining in a typical longwall working? How do the dynamics of caving alter with the variation of bulking factor, strata characteristics, and mining height? b. Discuss the supporting cycle and loading characteristics of the powered roof support deployed in the longwall operation. (Schematic is required)	[5+5]
3.	With neat sketches, describe the abutment stress distribution about the longwall panel.	[10]
4.	a. What is 'Roof Cavity'? Discuss the mechanism that leads to the formation of the roof cavity. (Schematic is required) b. Enumerate the mechanism of face damage and various forms of damage parameters. (Schematic is required)	[5+5]
	Section B - 30 Marks Instruction: Questions No. 5-6 are compulsory [2×15 = 30 Marks]	
5.	a. A longwall face of Mine 'A' is planned to operate under 350m cover depth with a face length of 200m and a face height of 3.5m. The physico-mechanical properties of the various roof layers are given in Table 1. Estimate the span of the main fall for the given mining condition using the relation: $W_s = 324 F_L^{-0.43} D^{-1.38} \sqrt{\frac{\sigma_{mIR} \times T_{IR}}{\gamma_{IR}}} \sqrt{\frac{\sigma_{mMR} \times T_{MR}}{\gamma_{MR}}}$ b. Following the same strata characteristics, if the mine management plans to operate the next face by changing the face length to 250m under the cover depth of 450m,	[8+5+2]

3.05 + 0.25 (5m)

	then what would be the span of the main fall for the new panel? c. In the given relation, why are we not considering the variable pertaining to coal strength and mining height for the determination of the main fall span in the given geo-mining condition?	
6.	a. A longwall face of Mine 'B' is planned to operate under the cover depth of 150m with a face length and face height of 180m and 3.5m, respectively. The Physico-Mechanical properties of overlying roof rocks related to the working are given in Table 2. Estimate the capacity of powered roof support that needs to be deployed in the given condition considering the canopy length of 4m, canopy width of 1.75m, and web depth of 0.8m for the working. Useful relations for reference to solve this question: $C_m = \left(\frac{A}{P} + 9.6h_f \right) + \left(\frac{kI_{\max}}{k_1 + 1.5} - 23 \right)$ b. Keeping the other parameters constant, If the mine management planned to operate the longwall face with a 4.5m face, then what would be the support capacity for altering the face height from 3.5m to 4.5m? c. If the $I_{\max}$ and thickness of cavable bed alter to 4915 and 31m, respectively, then what would be the required support capacity for this scenario? (Take face height as 3.5m)	[5+5+5]

Table 1: Physico-mechanical characteristics of the roof layers of Mine 'A'

Layers	Thickness (m)	Density (Kg/m ³)	RQD (%)	Compressive Strength (MPa)	Tensile Strength (MPa)
Coal	3.5	1400	40	23.16	1.54
Immediate Roof	35.23	2045	80	12.25	1.10
Main Roof	17.46	2050	76	12.89	1.13
Overburden	297.31	2089	74	13.21	1.26

Assume, $G = 0.03^\circ\text{C/m}$, $\beta = 8 \times 10^{-6} / ^\circ\text{m}$, $\nu = 0.25$, $\gamma = 0.023 \text{ MPa/m}$

$1.3 \times 10^4 \times 0.023$
 $\times (1.75 + 0.13)$

$\sigma_{\text{max}} / \text{mm}^2 = \left(\frac{\sigma_{\text{m}}}{1.6 \text{ m}^2} + \sigma_{\text{rock}} \right)$
 $\times 100$
 required factor

Table 2: Physico-mechanical properties of the overlying roof rocks above the coal seam for the longwall panel belonging to Mine 'B'

BED	Depth from surface, m		Rock Bed Characteristics	Depth above coal seam, m		Thickness of bed, t (m)	Density (Kg/m ³)	Weighted comp. $\Sigma K_i \sigma_{bi}$ (Kg/cm ²)	Weighted tens. $\Sigma K_i \sigma_{bi}$ (Kg/cm ²)	RQD, %	Average Length, L (cm)	Caving Index, I
	From	To		From	To							
Bed 14	110.65	118.24	Sandstone coarse grained	40.76	43.35	7.59	2348	261	41	96	25.48	6197
Bed 13	118.24	124.26	Sandy shale	34.74	40.76	6.02	1993	261	43	49	11.10	1422
Bed 12	124.26	125.24	Sandstone coarse grained	33.76	34.74	0.98	2317	212	34	78	22.75	1784
Bed 11	125.24	128.00	Sandstone medium grained	31.00	33.76	2.76	2382	306	49	76	21.70	4084
Bed 10	128.00	129.01	Sandstone medium grained	29.99	31.00	1.01	2382	306	49	96	33.67	2631
Bed 9	129.01	134.38	Alternate shale & sandstone	24.62	29.99	5.37	2392	586	71	88	14.58	5499
Bed 8	134.38	137.00	Sandstone fine grained	22.00	24.62	2.62	2378	350	52	82	23.00	4885
Bed 7	137.00	144.81	Alternate shale & sandstone	14.19	22.00	7.81	2369	335	50	83	18.92	3543
Bed 6	144.81	146.00	Sandstone medium grained	13.00	14.19	1.19	2382	306	49	99	59.00	3011
Bed 5	146.00	147.25	Alternate shale & sandstone	11.75	13.00	1.25	2373	320	48	96	25.00	1791
Bed 4	147.25	149.00	Sandstone medium grained	10.00	11.75	1.75	2382	306	49	90	15.82	3121
Bed 3	149.00	152.00	Sandstone coarse grained	7.00	10.00	3.00	2348	261	41	58	18.00	2176
Bed 2	152.00	153.82	Sandstone coarse grained	5.18	7.00	1.82	2336	243	38	95	22.75	2783
Bed 1	153.82	159.00	Grey shale	0.00	5.18	5.18	2367	341	50	94	18.78	3428

coal

0.72 x 50 x 1.5

2.05 + 0.25 (53)